TAKEONE · MATCH VIDEO | DECISION REPORT | 20 OCT 2026

Match Video Storage Decision

Where competition footage lives, how fast it plays, and what we do next.

Recommendation: C + YouTube highlights. Keep 20.7 GB per event forever instead of 46.3 GB, fit 386 events like 17 Oct on one 8 TB NAS, start videos in 1–2 s, and put only the 22 finals per event on YouTube. It scores 31 out of 35, the highest of the five options.

FOOTAGE PER CAMERA
380 min
76 bouts fought on 17 Oct

LIVE SERVER FILLS AFTER
0.7 of one event
63 GB free today, 96.9 GB needed

PERMANENT STORAGE SAVED
55 %
lighter encoding vs today

EVENTS ON ONE 8 TB NAS
386
vs 172 without lighter encoding

The five options

OPTION	WHAT IT MEANS	MAIN CATCH
Server only (today)	Everything stays on the live server's disk.	One big event fills the disk and takes the site down.
YouTube only	Upload every bout to YouTube and delete our copy.	Minors' bouts barely earn, quota allows ~6 uploads a day, a channel strike could wipe all footage.
B · NAS + server cache	Originals on a NAS; streaming versions on the server.	Works, but originals take twice the space of option C.
C · NAS + lighter encoding	As B, with modern quality-target encoding and a lighter archive copy.	No revenue on its own.
C + YouTube highlights	Option C, plus finals published to a YouTube playlist per event.	Needs guardian consent and Google's quota approval.

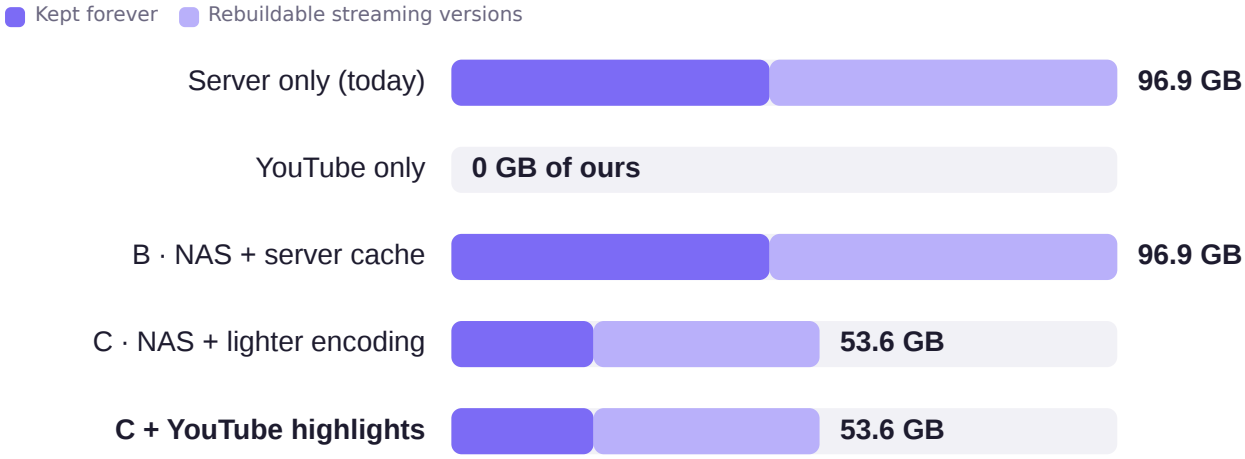
Decision: C + YouTube highlights

1. Disk guard on the live server this week.
2. A mirrored NAS beside the server for the originals, plus a second copy elsewhere.
3. Lighter, sharper video with a 1–2 second start.
4. Downloads that keep every point as a chapter.
5. YouTube for finals only, unlisted by default, our own copy always kept.
6. Live streams go from each phone straight to YouTube; our server and home line stay out of the path.

Storage

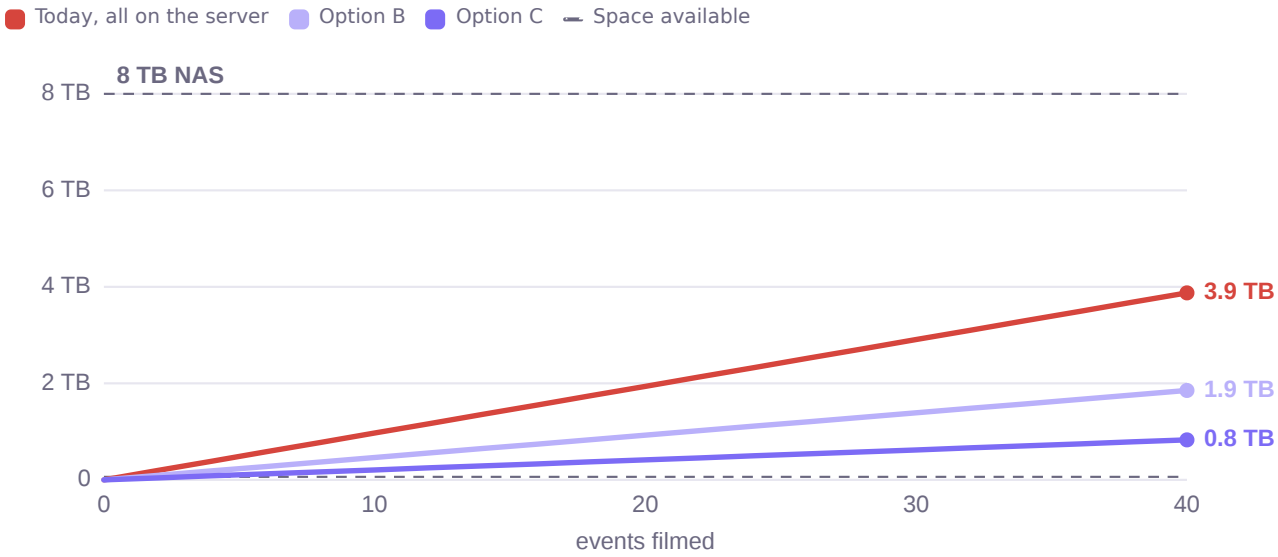
Storage per event, two cameras

Kept forever is the one copy we must never lose. Rebuildable is the streaming versions, which the server can delete and rebuild.



Permanent storage over a season

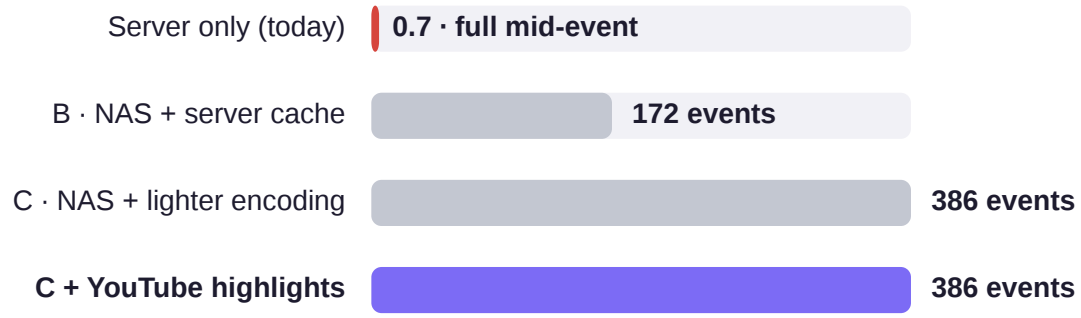
What piles up as events are filmed. Top dashed line: the 8 TB NAS. Bottom dashed line, just above zero: the 63 GB free on the live server today.



Capacity and speed

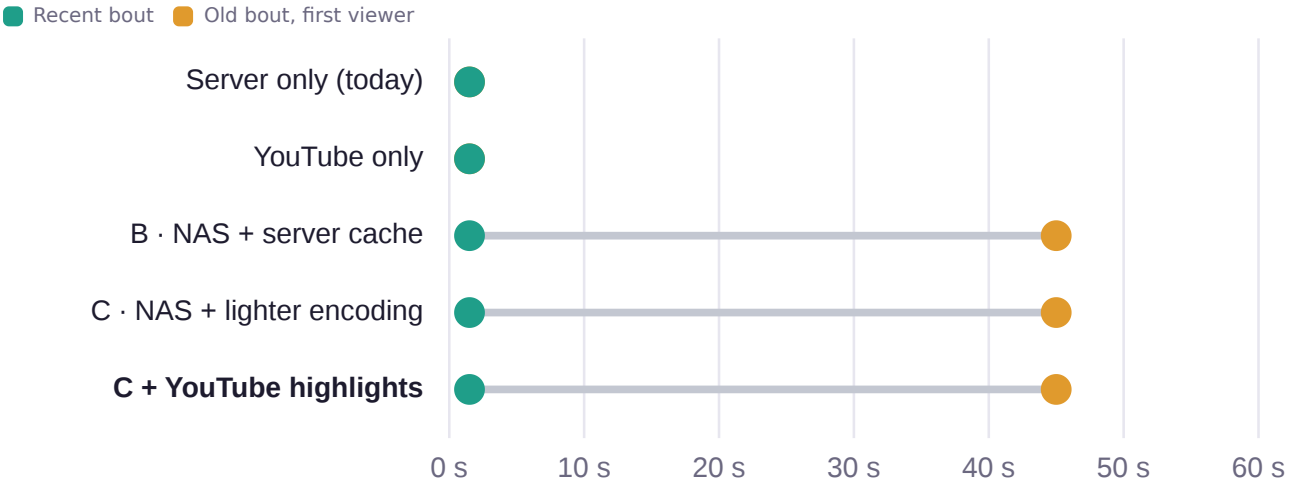
Events like 17 Oct that fit

On the space each option stores into. YouTube has no limit but keeps no copy of ours, so it is left out.



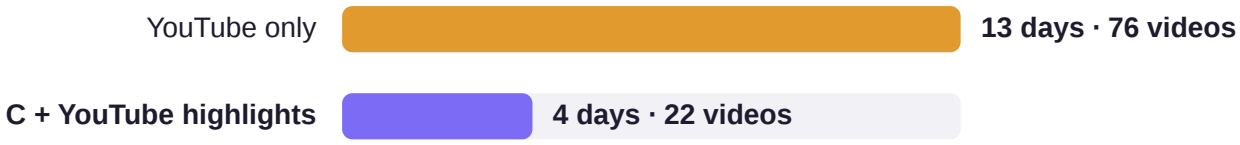
Seconds until the video starts

Recent = a bout still streamed from the server. Old = first viewer of an archived bout, estimate 30–60 s (middle shown). Serving the original while it rebuilds brings that to a few seconds.



Days to upload one event to YouTube

The default API quota allows about 6 uploads a day. Every bout takes about two weeks; only the finals take a few days.



Scorecard

How each option scores, 1 (poor) to 5 (best)

Our assessment from the discussion. Green is strong, red is weak.

Option	Storage cost	Playback speed	Footage safety	Privacy & minors	Revenue	Point replay	Hall with no internet	Total
Server only (today)	1	5	2	5	1	5	5	24/35
YouTube only	5	5	1	1	2	4	1	19/35
B · NAS + server cache	4	4	4	5	1	5	5	28/35
C · NAS + lighter encoding	5	4	4	5	1	5	5	29/35
C + YouTube highlights	5	4	4	4	4	5	5	31/35

Measured from the 17 Oct event

Bahrain Karate Premier League 2026. Every estimate starts from these.

110 bouts in the draw	**34** byes, not filmed	**22** divisions
4.7 median minutes a bout	**8** Mbit/s the camera records	**8.9** Mbit/s all streaming versions

How the numbers were worked out

- Bout counts and times are measured from the 17 Oct event's officiating log; free disk is measured on the live server today.
- Each fought bout gives about 5 minutes of video per camera; two cameras on the mat at 30 fps.
- Camera originals at the app's 8 Mbit/s setting; streaming versions are today's 480p, 720p and 1080p ladder.
- The lighter archive copy is about 3.5 Mbit/s H.265 or AV1; quality-target streaming saves about 35%.
- Start times are estimates; scores are a judgment, not a measurement.
- Only two real clips are stored so far, so no full event has been filmed end to end yet. Step 1 of the plan includes measuring the first real event.

ACTION PLAN · 17 STEPS · 6 PHASES

From today's disk to C + YouTube highlights

34 days
development, spread over the phases

4 tasks
for you: buy the NAS, a second-copy location, the Google application, a hall speed test per event

Every step
behind a flag or a setting, OFF until tested on stage

PHASE 1 Protect the live server

This week

#	STEP	WHY	WHO	EFFORT	SAFETY	DONE WHEN
1	**Disk guard** Refuse new camera uploads and alert the organiser when free disk falls below a set limit (e.g. 15 GB).	A full disk crashes the database and the whole site, not just video.	Dev	0.5 day	Flag, on after a stage test	A test upload is refused at the limit and the site stays up.
2	**Fast segment serving** Set MEDIA_INTERNAL_PREFIX and the matching nginx internal location, so nginx sends video segments instead of PHP.	The server sits on a home line; every bit of efficiency counts once several people watch at once.	Dev	0.5 day	Config only; unset to roll back	Segments return via X-Accel-Redirect; access is still checked per segment.

PHASE 2 The NAS

After the hardware arrives

#	STEP	WHY	WHO	EFFORT	SAFETY	DONE WHEN
3	**Buy and install the NAS** 2-bay NAS, 2 × 8 TB drives mirrored, on the same local network as the live server.	Mirroring means one dead drive loses nothing.	You	1 day	Hardware	NAS reachable from the server over SMB or a mount.
4	**Attach it as a media vault** Through the existing Storage admin page. No new code.	The vault system already supports SMB and mounted storage.	Dev	0.5 day	Admin setting	A test file moves to the NAS and plays back.
5	**Auto-archive originals** Once a clip has its streaming versions, move the original to the NAS; delete the local copy only after the hash check passes.	Frees the server disk event by event.	Dev	2 days	Flag, default OFF	Originals land on the NAS with matching hashes; playback unchanged.
6	**Second copy of the originals** Point BACKUP_MEDIA_DEST at a second disk or cheap cloud storage.	Otherwise the NAS holds the only copy of the competition footage.	Dev + You	0.5 day	Config	Nightly media backup runs and verifies.

PHASE 3 Lighter, sharper video

After Phase 2

#	STEP	WHY	WHO	EFFORT	SAFETY	DONE WHEN
7	**Quality-target H.265 ladder** Encode to a quality target on the RTX 3060 (hevc_nvenc), keep a small H.264 fallback, 2-second opening segments, start on the smallest quality.	About 30–40% smaller at the same sharpness; 1–2 s start.	Dev	3 days	Flag, default OFF; old ladder stays	Same bout encoded both ways; new one smaller, plays on Android, iPhone and the app.
8	**Lighter archive copy** Re-encode each camera original once to about 3.5 Mbit/s H.265 or AV1.	Cuts permanent storage from 46.3 to 20.7 GB per event.	Dev	2 days	Flag, default OFF	Decision needed: keep the 8 Mbit/s original or not.
9	**Drop unwatched streaming versions** Delete streaming versions not watched in N days; rebuild from the NAS on the next play, serving the original meanwhile.	Keeps only recent and popular events on the server disk.	Dev	3 days	Flag, default OFF	An old bout starts in a few seconds and is fully rebuilt in under a minute.

PHASE 4 Downloads that keep the points

In parallel with Phase 3

#	STEP	WHY	WHO	EFFORT	SAFETY	DONE WHEN
10	**MP4 download with point chapters** Download the bout as an MP4 with every point as a chapter marker (e.g. "0:42 Red, Yuko").	People keep the bout and can still jump to each point.	Dev	2 days	Flag; same access rules as watching	Chapters show in VLC and phone players; minors follow the viewing rules.
11	**Burned-in scoreboard (optional)** A running scoreboard drawn on the picture, built on request and deleted after a day.	Works in any player.	Dev	3 days	Flag, default OFF	Decision needed: build it or chapters only.

PHASE 5 YouTube for finals

Start the paperwork now

#	STEP	WHY	WHO	EFFORT	SAFETY	DONE WHEN
12	**Google project and quota request** Create the Google Cloud project, the channel and apply for more API quota.	Default quota is about 6 uploads a day; approval takes weeks.	You	2–6 weeks wait	External	Quota approved; channel verified.
13	**YouTube highlights module** Its own module: connect the channel, one playlist per event, organiser picks which finals, unlisted by default, guardian consent before a minor is uploaded, MCP tools.	Up to 22 videos per event for publicity and revenue; our own copy always stays.	Dev	5 days	Flag, default OFF	One test event publishes its finals playlist; point replay works on the embed.

PHASE 6 Live to YouTube from the phone

After step 12 is approved

#	STEP	WHY	WHO	EFFORT	SAFETY	DONE WHEN
14	**Hall internet check** Test the hall upload speed before each event: about 4–6 Mbit/s per mat that goes live.	The stream leaves from the hall, so the hall line is the limit.	You	15 min per event	Checklist	Speed test logged for the hall before the event.
15	**Live stream in the camera app** The phone keeps recording 1080p for our archive and also sends a 720p stream straight to YouTube. Our server and the home line are never in the path.	YouTube serves every live viewer, so a big audience costs us nothing.	Dev	5 days	Flag, default OFF; recording never stops if the stream drops	One mat streams for an hour while recording; the bout clips still upload afterwards.
16	**Live broadcast per mat** Create a YouTube live broadcast for each mat of an event, started and stopped from the existing on/off switch in the event console, unlisted by default, embedded on the event page.	The organiser controls what goes out, from the scoring table.	Dev	3 days	Flag, default OFF; guardian consent before the event	The event page shows the live player for each mat; switching off ends the broadcast.
17	**Live scoreboard on the stream** The phone draws a small scoreboard onto its stream from the scores it already receives.	Viewers see every point as it happens.	Dev	4 days	Flag, default OFF	Points appear on the live stream within a second of being scored.

Decisions we need from you

1 · Where the NAS sits
Next to the live server (fast) or elsewhere over the internet (slow for old bouts).

2 · Keep the 8 Mbit/s original?
Keeping it doubles permanent storage; the lighter copy looks the same for this footage.

3 · Burned-in scoreboard?
Works in any player but re-encodes per download; chapters only are instant.

4 · YouTube channel owner
Which account owns the channel, and who signs off guardian consent for minors, live and recorded.

5 · How many mats go live
Each live mat needs a phone on a charger and about 4–6 Mbit/s of hall upload.

6 · Home line limit
Your residential line holds about 5–8 people watching replays in HD at once. Big audiences belong on YouTube.

Safety rules for every step

- Nothing that works today is removed; new paths run beside the old ones until verified.
- A verified database backup before every migration.
- No original is deleted until its copy on the NAS passes a hash check.
- Every video endpoint checks who may watch on every request, including downloads.
- A live stream failing never stops the recording; the archive copy always comes first.
- Each step ships with its MCP tool and tests, and is tried on stage before live.

Effort figures are development estimates. Storage figures are estimates from measured bout data; see the scorecard page.